

Schwarz Rearrangement Can Increase a Thresholded Nonlocal Energy

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Status: counterexample, subject to human review. For $N = 1$, $p = 3/2$, and $\delta = 1$, an explicit nonnegative four-level step function satisfies $I_1(u^*) - I_1(u) > 43/1000$. Thus the universal Schwarz-rearrangement inequality asked in Open Problem 2.2 of arXiv:1704.08077 is false.

1 Source problem

Nguyen and Squassina study, for $\delta > 0$, the functional

$$I_\delta(v) = \iint_{\{|v(y)-v(x)|>\delta\}} \frac{\delta^p}{|x-y|^{N+p}} dx dy \quad (1)$$

and prove that it can increase under a single polarization. They then ask on arXiv PDF page 5 of arXiv:1704.08077v2 [1]:

Source Open Problem 2.2. *For $N \geq 1$, is $I_\delta(u^*) \leq I_\delta(u)$ for every measurable $u : \mathbb{R}^N \rightarrow \mathbb{R}^+$ and every $\delta > 0$?*

In the above example, the function U is not non-negative and H does not contain the origin. However, this point can be handled similarly as in the case $N = 1$.

The following question remains open:

Open problem 2.2. Let $N \geq 1$. It is true that $I_\delta(u^*) \leq I_\delta(u)$ for any measurable $u : \mathbb{R}^N \rightarrow \mathbb{R}^+$ and $\delta > 0$?

The exponent p is part of the preceding definition and is suppressed in the notation. A single choice of N, p, δ, u with the reverse strict inequality disproves the universal formulation.

2 Idea

The threshold in (1) is strict. We therefore use a step function whose adjacent levels differ by exactly $\delta = 1$. The singular kernel is never integrated across a touching pair of regions whose value gap is active, so the energy remains finite. The levels 3 and 4 form a high-value core inside a long level-2 buffer. Moving that core off center does not change the distribution of values, but it lowers the sum of the long-range interactions selected by the nonconvex threshold. Schwarz rearrangement moves the core back to the center and increases the energy.

3 Counterexample

Theorem 1. For $N = 1$, $p = 3/2$, and $\delta = 1$, there is a nonnegative measurable $u : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$I_1(u^*) > I_1(u).$$

More precisely, the difference is greater than $43/1000$.

Proof. Endpoint values below are immaterial. Define

$$u(x) = \begin{cases} 4, & x \in [8, 10], \\ 3, & x \in [7, 8] \cup [10, 11], \\ 2, & x \in [-14, 7] \cup [11, 14], \\ 1, & x \in [-15, -14] \cup [14, 15], \\ 0, & \text{otherwise.} \end{cases}$$

Its superlevel measures are

t range	$[0, 1)$	$[1, 2)$	$[2, 3)$	$[3, 4)$
$ \{u > t\} $	30	28	4	2.

Consequently its one-dimensional Schwarz rearrangement is

$$u^*(x) = \begin{cases} 4, & x \in [-1, 1], \\ 3, & x \in [-2, -1] \cup [1, 2], \\ 2, & x \in [-14, -2] \cup [2, 14], \\ 1, & x \in [-15, -14] \cup [14, 15], \\ 0, & \text{otherwise.} \end{cases}$$

Only pairs of levels differing by at least two contribute to I_1 . Such regions are separated by a positive distance, so both energies are finite. For disjoint intervals $[a, b]$ and $[c, d]$ with $b < c$, direct integration gives

$$\int_a^b \int_c^d (y - x)^{-5/2} dy dx = \frac{4}{3} \left(\frac{1}{\sqrt{c-b}} + \frac{1}{\sqrt{d-a}} - \frac{1}{\sqrt{c-a}} - \frac{1}{\sqrt{d-b}} \right). \quad (2)$$

If $d = \infty$, the two terms involving d are simply omitted. The left exterior half-line is handled by reflection. Summing twice (2) over the unordered pairs of regions whose values differ by more than one yields

$$\begin{aligned} I_1(u) &= \frac{112}{15} + \frac{2\sqrt{2}}{3} + \frac{2\sqrt{6}}{3} + \frac{4\sqrt{26}}{39} + \frac{8\sqrt{21}}{63} - \frac{8\sqrt{3}}{9} - \frac{8\sqrt{7}}{21} - \frac{8\sqrt{22}}{33} - \frac{16\sqrt{29}}{87}, \\ I_1(u^*) &= \frac{28}{3} + \frac{16\sqrt{15}}{45} + \frac{16\sqrt{17}}{51} - \frac{8\sqrt{3}}{9} - \frac{32\sqrt{13}}{39} - \frac{16\sqrt{29}}{87}. \end{aligned}$$

Thus, after cancelling the common terms,

$$\begin{aligned} \Delta := I_1(u^*) - I_1(u) &= \frac{28}{15} - \frac{32\sqrt{13}}{39} - \frac{2\sqrt{6}}{3} - \frac{2\sqrt{2}}{3} - \frac{8\sqrt{21}}{63} - \frac{4\sqrt{26}}{39} \\ &\quad + \frac{8\sqrt{7}}{21} + \frac{8\sqrt{22}}{33} + \frac{16\sqrt{17}}{51} + \frac{16\sqrt{15}}{45}. \end{aligned} \quad (3)$$

For a fully rational sign certificate, use the lower bounds

$$\sqrt{7} > 2.64575, \quad \sqrt{22} > 4.690415, \quad \sqrt{17} > 4.123105, \quad \sqrt{15} > 3.872983$$

in the positive terms of (3), and the upper bounds

$$\sqrt{13} < 3.605552, \quad \sqrt{6} < 2.449490, \quad \sqrt{2} < 1.414214, \quad \sqrt{21} < 4.582576, \quad \sqrt{26} < 5.099020$$

in its negative terms. Each bound is verified by squaring. Substitution gives the exact rational estimate

$$\Delta > \frac{1376155019}{31906875000} > \frac{43}{1000} > 0.$$

This proves the reverse strict inequality. □

4 Checks, scope, and novelty

The construction is genuinely measurable and nonnegative. Its jumps are not a hidden source of infinite energy: every touching pair of constant regions differs by at most one, whereas the defining condition is $|u(x) - u(y)| > 1$. Formula (2) therefore accounts for every nonzero contribution. The accompanying `verification.py` independently enumerates all contributing pairs and verifies both the numerical difference and the rational lower certificate.

Theorem 1 answers Open Problem 2.2 negatively under its universal reading, at the explicit exponent $p = 3/2$. It does not settle the paper’s separate Open Problem 4.1 concerning Pólya–Szegő for distributional Riesz fractional gradients. Nor does it classify which other fixed exponents or higher dimensions satisfy or fail the inequality.

Before promotion, the four lightweight run indexes were searched by exact arXiv id, title, and the functional/rearrangement keywords. A bounded search through 12 August 2026 used the exact title and combinations of “ $I_\delta(u^*)$ ”, “Schwarz rearrangement”, “threshold”, and “nonlocal functional”. It found the source and papers on different rearrangement energies, but no later proof or counterexample for Open Problem 2.2. The journal landing page’s cited-by list likewise exposed only unrelated applications. This is a bounded novelty check, not a claim of exhaustive bibliographic priority.

References

- [1] H.-M. Nguyen and M. Squassina, *Some remarks on rearrangement for nonlocal functionals*, *Nonlinear Analysis* **162** (2017), 1–12; arXiv:1704.08077v2.